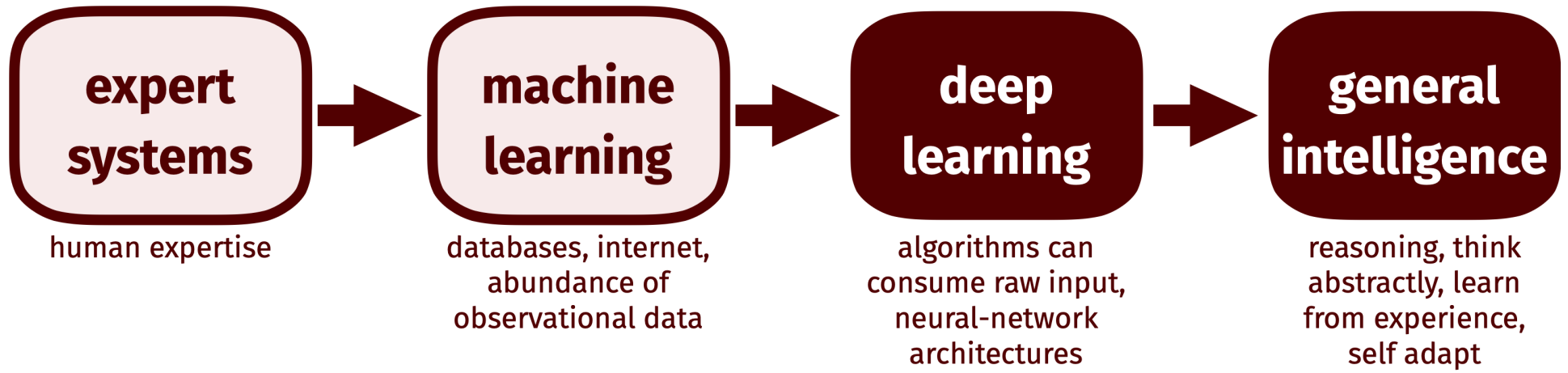




Deduction

Theory
↓
Hypothesis
↓
Observation
↓
Confirmation



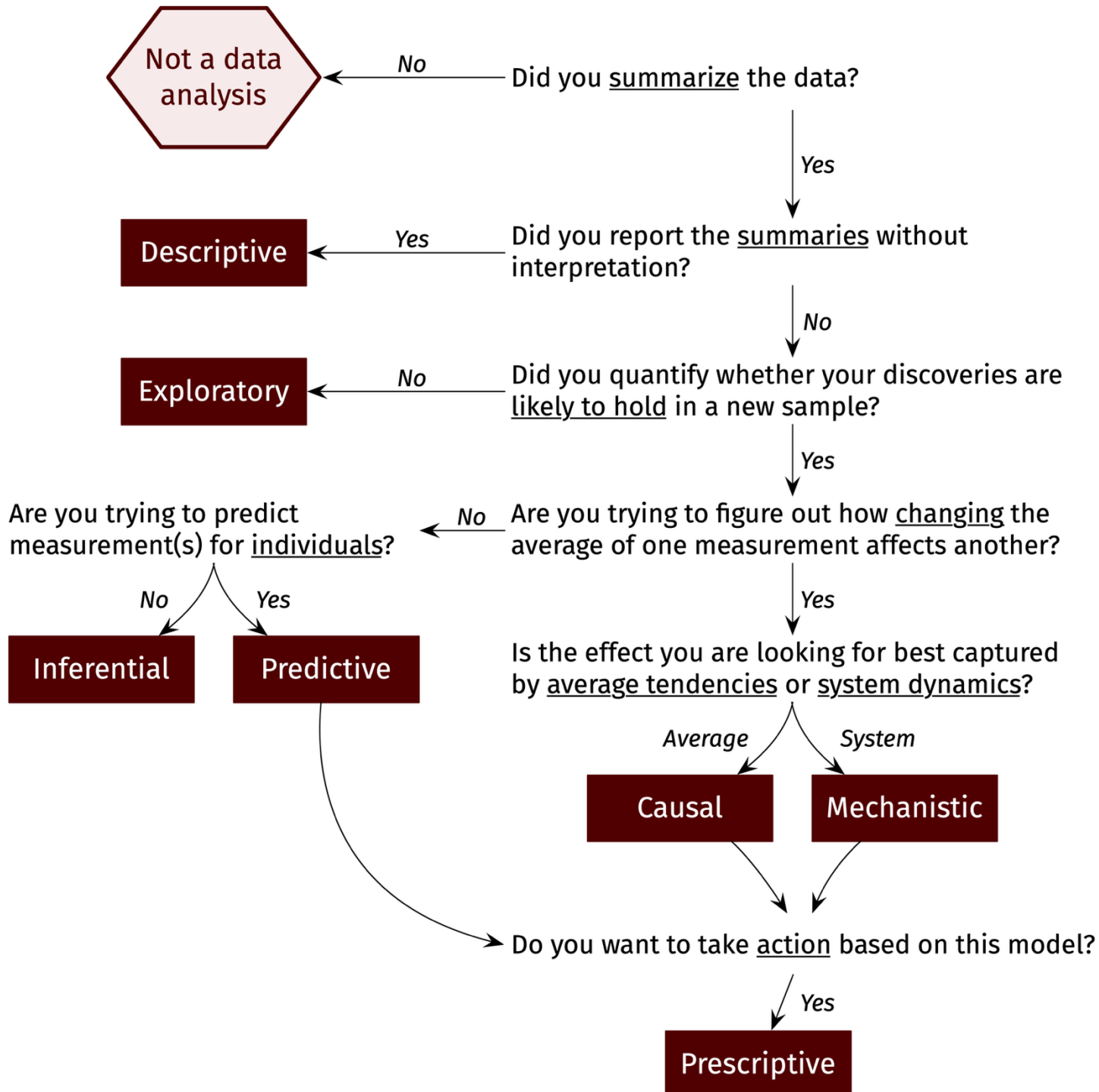
Aristotle

Induction

Observation
↓
Pattern
↓
Hypothesis
↓
Theory



Sherlock



DAEN 300	Topic	Main coding experience	Book
1	Computational thinking & getting started with Python	Run programs; IDE/notebook; expressions; simple scripts	Ch. 1–2
2	Variables, types, expressions & input/output	Numeric calculations, strings, keyboard input, formatted output	Ch. 2
3	Decisions and iteration	if/elif/else, while, for, range; build small interactive programs	Ch. 2
4	Computational numerical methods	Exhaustive enumeration, approximation, bisection search, Newton–Raphson	Ch. 3
5	Functions and abstraction	Parameters, return values, scope, specifications, decomposition	Ch. 4
6	Lists, tuples & sequences	Indexing, slicing, mutation, iteration, list comprehensions	Ch. 5
7	Dictionaries, sets & data organization	Dictionaries, sets, counting, lookup, structured records	Ch. 5
8	Recursion and problem decomposition	Fibonacci, palindrome-type problems, recursive thinking	Ch. 6
9	Modules, libraries & files	Imports, packages, reading/writing files, processing real data	Ch. 7
10	Testing, debugging & error handling	Test cases, black-/glass-box testing, debugging, exceptions, assertions	Ch. 8–9
11	Object-oriented programming	Classes, objects, methods, inheritance, encapsulation	Ch. 10
12	Algorithms, efficiency & data structures	Complexity, linear/binary search, sorting, merge sort, hash tables	Ch. 11–12
13	Data visualization & Semester 1 project	Matplotlib; combine files, classes, algorithms and visualization	Ch. 13

DAEN 301	Topic	Main coding experience	Book
1	Optimization and greedy algorithms	Knapsack-type problems; implement and compare optimization strategies	Ch. 14
2	Graphs and graph algorithms	Graph representation, DFS, BFS, shortest paths	Ch. 14
3	Dynamic programming	Memorization, Fibonacci, knapsack, divide-and-conquer comparisons	Ch. 15
4	Random walks & computational models	Build classes/models representing random movement; visualize results	Ch. 16
5	Probability through programming	Randomness, probability experiments, distributions	Ch. 17
6	Monte Carlo simulation	Repeated trials, estimation, simulation design and performance	Ch. 18
7	Sampling, confidence & uncertainty	Sampling experiments, CLT, standard error, confidence	Ch. 19
8	Experimental data & regression	Fit models, linear regression, (R^2), model evaluation	Ch. 20
9	Hypothesis testing & Bayesian reasoning	Randomized trials, significance, conditional probability, Bayes	Ch. 21
10	Responsible interpretation of data	Detect misleading graphs, bias, extrapolation and statistical mistakes	Ch. 22
11	Practical data analysis with Pandas	CSV, Series, DataFrames, filtering, grouping, manipulation	Ch. 23
12	Machine learning fundamentals & clustering	Feature vectors, distance metrics, k-means, unsupervised learning	Ch. 24–25
13	Classification & final data-science project	k-NN, regression classifiers, classifier evaluation; complete ML pipeline	Ch. 26